

a)

$$\hat{\beta}_1 = \frac{\hat{\sigma}_{y,x_1}}{\hat{\sigma}_{x_1}^2} \xrightarrow[n \rightarrow \infty]{\text{converges in probability}} \frac{\sigma_{y,x_1}}{\sigma_{x_1}^2} = \frac{\text{Cov}(y, x_1)}{\text{Var}(x_1)} = \frac{\text{Cov}(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, x_1)}{\text{Var}(x_1)} =$$

$$\frac{\cancel{\text{Cov}(\beta_0, x_1)} + \text{Cov}(\beta_1 x_1, x_1) + \text{Cov}(\beta_2 x_2, x_1) + \cancel{\text{Cov}(u, x_1)}}{\text{Var}(x_1)} =$$

covariance of a constant = 0 exogeneity means this term is 0

$$\frac{\text{Cov}(\beta_1 x_1, x_1) + \text{Cov}(\beta_2 x_2, x_1)}{\text{Var}(x_1)} = \frac{\beta_1 \cdot \text{Cov}(x_1, x_1) + \beta_2 \cdot \text{Cov}(x_2, x_1)}{\text{Var}(x_1)} =$$

$$\beta_1 + \frac{\beta_2 \cdot \text{Cov}(x_2, x_1)}{\text{Var}(x_1)} = \beta_1 + \frac{\beta_2 \cdot \sigma_{x_1, x_2}}{\sigma_{x_1}^2}$$

b) $\hat{\beta}_1$ is not consistent, because it does not converge in probability to β_1 .

The asymptotic bias is $\left(\beta_1 + \frac{\beta_2 \cdot \sigma_{x_1, x_2}}{\sigma_{x_1}^2} \right) - \beta_1 = \frac{\beta_2 \cdot \sigma_{x_1, x_2}}{\sigma_{x_1}^2}$

c) $\beta_0 = 1 \quad \beta_1 = 5 \quad \beta_2 = 4 \quad \sigma_{x_1}^2 = 1 \quad \sigma_{y, x_1} = 4.2 \quad \sigma_{y, x_2} = 3 \quad \sigma_{x_1, x_2} = -0.2$

$$\beta_1 = 5 \quad \frac{\beta_2 \cdot \sigma_{x_1, x_2}}{\sigma_{x_1}^2} = \frac{4 \cdot (-0.2)}{1} = -0.8$$

$\hat{\beta}_1$ underestimates the true parameter, β_1 , on average, because the asymptotic bias is negative.